Environmental Product Declaration

In accordance with ISO 14025:2006 and EN 15804:2012+A2:2019/AC:2021

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Jolywood

N-type Bifacial Double Glass Photovoltaic Modules

JW-HD144N-182
JW-HD156N-182

Note: An EPD should provide current information and may be updated if conditions change. The stated validity is therefore subject to the continued registration and publication at www.environdec.com

EPD of multiple products, based on a representative product.

Programme information

Programme:	The International EPD® System EPD International AB Box 210 60 SE-100 31 Stockholm Sweden www.environdec.com info@environdec.com

Product Category Rules (PCR)	
CEN standard EN 15804 serves as the Core Product Category Rules (PCR)	
Product category rules (PCR):	PCR 2019: 14 PCR Construction products v1.3.1 c-PCR-016 Photovoltaic modules and parts thereof (adopted from EPD Norway 2022-04-27)
PCR review was conducted by:	The Technical Committee of the International EPD® System. A full list of members available on www.environdec.com. The review panel may be contacted via info@environdec.com Chair of the PCR review: No appointed chair
EPD Author:	TÜV SÜD Certification and Testing (China) Co., Ltd. Shanghai Branch
Independent third-party verification of the declaration and data, according to ISO 14025:2006: ☐ EPD process certification ☒ EPD verification	
Third party verifier:	Marcel Gómez Ferrer – Marcel Gomez Environmental Consulting – Info@marcelgomez.com
Approved by:	*The International EPD® System*
Procedure for follow-up of data during EPD validity involves third party verifier: ☐ Yes ☒ No	

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Company information

Owner of the EPD:
Jolywood (Taizhou) Solar Technology Co., Ltd.
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Address: Kaiyang Road, Jiangyan Economic Development Zone, Taizhou, Jiangsu, China

Jolywood (Jiangsu) Light Energy Technology Co., Ltd.
Address: No. 6-1 Kaiyang Road, Jiangyan Economic Development Zone, Taizhou, Jiangsu, China

Description of the Company:
Founded in 2008, Jolywood is a national high-tech enterprise. It was successfully listed in 2014 (stock code: 300393). The company focuses on innovative R&D and high-quality manufacturing of photovoltaic auxiliary materials, solar cell and module, system integration and other products. As the world largest N-type bifacial solar cell and module manufacturer with 6.6GW production capacity, and world largest backsheet manufacturer with 30% market share, Jolywood was awarded financial health Top3 by Photon. Jolywood is the world's largest manufacturer of N-type TOPCon bifacial solar panel, with a registered capital of 1.5 billion CNY, N-type TOPCon bifacial solar panel of more than 3GW capacity. In the N-type TOPCon bifacial solar panel process, Jolywood has complete independent intellectual property rights. Up to now, it has applied for 99 related patents, including 58 granted patents and PCT5 items and maintain long-term research cooperation with the international renowned IMEC Institute, Sun Yat-sen University, Shanghai Jiao Tong University.

Product information

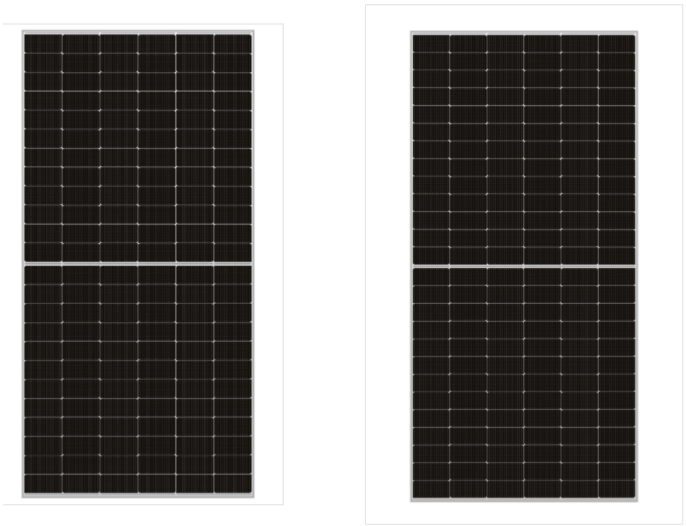
Product name:
Jolywood N-type Bifacial Double Glass Photovoltaic Modules
JW-HD144N-182
JW-HD156N-182

UN CPC code:
461 Electric motors, generators and transformers, and parts thereof

Geographical scope: Global

Product description and application:
Jolywood N-Type bifacial modules can be widely used in rooftop and ground solar farms. Also, the products have wider application areas, such as high temperature regions, heavy snow regions, water surface systems, agriculture greenhouses, etc. With higher efficiency, higher bifaciality (up to 85%), lower degradation, and lower temperature co-efficiency (≤0.32%), the N-Type solar systems can generate 10-30% extra power than p-type mono facial products, therefore to reduce solar projects LCOE, bring better investment return for customers. Bifacial modules combine leading NTOPCon technology, 11BB and half-cell. The Jolywood N-type Bifacial Half-cell Module can reach

power output up to 630W. N-type material has zero LID/LeTID risk, and make modules to be higher reliable, higher bifaciality, higher efficiency, lower temperature coefficient and longer lifetime.



JW-HD144N-182 JW-HD156N-182

Figure 1 Jolywood's PV module

Product identification:

Table 1 Product technical specifications

Series (brand name)	JW-HD144N-182	JW-HD156N-182
Power output range (W)	550~585	605~630
Dimensions(mm^3)	2278*1134*30	2465*1134*30
Area (m^2)	2.583	2.795
Module efficiency (%)	22.45	22.53
Weight (kg)	32.50	34.50
Weight (incl. package)	32.643	34.646
First year degradation (%)	1	1
Annual degradation (%)	0.4	0.4
Type of cell/technology	Mono-Si/N-ToPCON	Mono-Si/N-ToPCON
Cells configuration	6*12+6*12	6*13+6*13
Reference service life (years)	25	25

Content declaration

Raw materials of the different PV modules are mostly the same, including solar cells, solar glass, frame, silicone gel, junction box and packaging etc. The compositions of raw materials are listed in Table 2 and Table 3.

Table 2 Raw materials compositions- JW-HD144N-182

Product components	Weight, kg	Post-consumer recycled materials, weight -%	Biogenic material, weight-% and kg C/kg
Glass	26.428±10%	0	-
Frame	2.3±10%	0	-
EVA	2.194±10%	0	-
Solder	0.242±10%	0	-
Cell	0.778±10%	0	-
Junction box	0.25±10%	0	-
Silicone gel	0.29±10%	0	-

Flux	0.016±10%	0	-
Sum	**32.5**	**0**	**-**
Packaging materials	**Weight, kg**	**Weight -% (versus the product)**	**Weight biogenic carbon, kg C/kg**
Wood pallet	1.029±10%	3.17	4.72E-1
Corrugated box	0.341±10%	1.05	4.50E-1
Packaging paper	0.085±10%	0.26	4.17E-1
Packaging bag	0.029±10%	0.09	0
Packaging film	0.029 ±10%	0.09	0

Table 3 Raw materials compositions- JW-HD156N-182

Product components	Weight, kg	Post-consumer recycled materials, weight -%	Biogenic material, weight-% and kg C/kg
Glass	27.252±10%	0	-
Frame	2.88±10%	0	-
EVA	2.608±10%	0	-
Solder	0.244±10%	0	-
Cell	0.9±10%	0	-
Junction box	0.25±10%	0	-
Silicone gel	0.349±10%	0	-
Flux	0.017±10%	0	-
Sum	**34.5**	**0**	**-**
Packaging materials	**Weight, kg**	**Weight -% (versus the product)**	**Weight biogenic carbon, kg C/kg**
Wood pallet	1.378±10%	3.99	4.72E-1
Corrugated box	0.333±10%	0.97	4.50E-1
Packaging paper	0.083±10%	0.24	4.17E-1
Packaging bag	0.032±10%	0.09	0
Packaging film	1.378±10%	3.99	0

No substance in the product greater than 0.10% by weight is present on the "List of Potentially Hazardous Substances" candidates for authorization under the REACH legislation.

<u>Manufacturing Process:</u>

The manufacturing process of PV modules is illustrated in Figure 2.

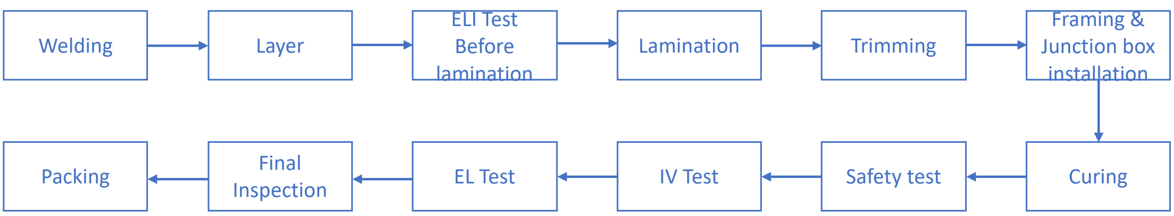


Figure 2 Manufacturing process

<u>Step 1: Welding</u>

The bus bar is welded to the main grid line on the front and back of the cell. The bus bar is tinned copper strip. The welding machine can spot-weld the strip on the main grid line in the form of multiple points.

<u>Step 2: Layer</u>

Connect the cells in series to form a PV layer. The positioning of the cell mainly depends on the template.

Step 3: Electroluminescent (EL) Imaging test before lamination
Conduct appearance and EL imaging inspection on the PV modules before lamination.

Step 4: Lamination
The assembled PV modules are put into the laminating machine, the air in the modules is extracted by vacuuming, and then the Ethylene Vinyl Acetate (EVA)/ Polyolefin Encapsulant (POE) is melted by heating, and the battery, glass and backplane are bonded together.

Step 5: Trimming
During lamination, EVA melts and extends outward due to pressure to form burr, so it should be cut off after lamination.

Step 6: Framing & junction box installation
Aluminum frame is installed on the glass module to increase the strength of the module, further seal the battery module and prolong the service life of the Cell. Weld a box at the back lead of the module to facilitate the connection between the battery and other devices or batteries.

Step 7: Curing
Under the condition of 25 ± 5 °C and more than 60% humidity, curing the PV Modules frame glue and junction box glue.

Step 8: Safety test
A certain voltage is applied between the frame and the electrode lead to test the withstand voltage and insulation strength of the module to ensure that the module will not be damaged under adverse natural conditions (such as lightning strike). Test the grounding performance of modules to ensure the safety of modules.

Step 9: IV Test
standard test condition (STC), the output power of the module is calibrated and its output characteristics are tested.

Step 10: EL Test
Electroluminescent imaging technology is used to detect the potential defects and control the product quality.

Step 11: Final inspection
Inspect the appearance of the modules and judge its grade.

Step 12: Packing
Package the modules after testing according to the standard.

Distribution Process:

Table 4 Information on the distribution stage

Vehicles	Capacity utilization (incl. return) %	Type of vehicle	Distance km	Fuel/energy consumption per tkm (kg/tkm)	Fuel/energy consumption per km (kg/km)
Truck	36.7	EURO6 16-32 ton	1141.9	Diesel: 0.0366	1.19E-4~1.26E-4
Boat	70	Container ship	8646.5	Heavy oil: 0.0025	8.12E-6~8.62E-6

Note: Range of fuel consumption per km (kg/km) due to different product weights

Product installation:

The PV module can be installed both as rooftop and ground-mounted type. According to the product category rules (PCR), mounting structures like inverter, wiring, etc., are not considered in this study. The electricity and diesel consumption have been derived from the ecoinvent database process for a photovoltaic plant installation with a capacity of 570 kWp, and these values have been downscaled to the power output of different modules analyzed in this study. Waste generation and treatment of packaging materials are considered in this stage. The waste generated from the product packaging, mainly consisting of waste wood pallets, is accounted for in this stage, transportation of waste is assumed as 200km. The treatment of the waste wood pallets is modeled as 75% recycling and 25% incineration. Other packaging materials, including paper and plastic film, are modeled as 100% incineration.

Use & Maintenance:

For the use stage (B1) of the PV products, no energy and materials inputs, or emissions are involved. As for the maintenance stage (B2), water used for cleaning to maintain the performance is considered, 0.23L water used per module each time, and 2 times in a year are assumed. During the operation of PV module, no repair (B3), replacement (B4), and refurbishment (B5) is required.

It is assumed that there is no operational electricity (B6) or water consumption (B7). To calculate the expected energy production over the lifetime of the panels, the following formula may be used:

$$E_1 = S_{\mathrm{rad}} * A * y * PR * (1 - \mathrm{deg})$$

Where:

E_1= Energy produced in the first year of operation, kWh/year

S_{rad} = Site specific annual average solar radiation on module (shadings not included), kWh/kWp/year. The annual radiation must take into consideration the specific inclination (slope, tilt) and orientation.

A = Area of module, m^2.

y = Module yield: electrical power, kWp for standard test conditions (STC) of the module divided by the area of the module.

STC: The ratio is given for standard test conditions: irradiance 1000 W/m^2, cell temperature 25 °C, wind speed 1 m/s, AM1.5.

PR = Performance ratio, coefficient for losses. Site specific performance ratio can be modelled with PV simulation software tools, such as PVSYST or similar.

Energy production second year of operation:

$$E_2 = E_1 * (1 - \mathrm{deg})$$

Energy production n year of operation:

$$E_n = E_1 * (1 - \mathrm{deg})^{n-1}$$

Energy production over reference service life of module, assuming linear annual degradation:

$$E_{RSL} = E_1 * (1 + \sum_{n=1}^{RSL-1} (1 - deg)^n)$$

End-of-Life:

For end-of-life (EoL) stage, assumptions are made due to a lack of data. Decommissioning stage (C1) of PV modules is assumed to be taken with same energy and fuel consumption as for installation stage. Transportation distance from the plant site to the waste treatment site (C2) is assumed to be 50km. Waste processing (C3) stage is assumed to be mechanically treated to yield the bulk materials with an electricity consumption of 0.277kWh/kg module, based on data from the IEA.

This study refers to legal requirements issued by Waste Electrical and Electronic Equipment (WEEE) under the EU scenario. The required recycling rate for waste PV modules is 85% according to 2012/19/EU-Article 11 & ANNEX V. According to the IEA report, the recycling rate of PV modules is above 85% in the main EU countries. Therefore, it is safe to use 85% as a recycling rate for waste PV modules. 15% of the waste components (cells, glass, and waste plastics) end up to disposal stage (C4). The plastic will be sent to incineration, while the cell and unrecovered glass will be treated as inert materials for landfilling. In this study, bulk materials (glass, aluminium, and copper) are recovered, while plastic components will be incinerated for energy recovery. It is assumed that 100% metal components and 85% glass will be recycled. 15% of contaminated glass and other unrecyclable materials will go to landfill, while the plastic components will go to incineration.

Benefits and loads beyond the system boundary:

100% metal scrap (aluminium, copper, and silver) and 85% of glass scrap will be recycled. The plastic components are incinerated with energy recovery. Efforts required by secondary production, loss of materials and quality are considered.

LCA information

The study is developed according to ISO 14040/14044, and follows the International EPD® System: *PCR 2019: 14 PCR Construction products v1.3.1, C-PCR-016 Photovoltaic modules and parts thereof (adopted from EPD Norway 2022-04-27),* and General Programme Instructions (GPI) v4.0.

Functional unit:

1 Wp of manufactured photovoltaic module, with activities needed for a study period for a defined reference service life (≥80% of the labelled power output, estimated at 25 years). The converting factor to convert the results related to the functional unit to declared unit (1 m^2 PV module) is 226.5 Wp/m^2 for JW-HD144N and 225.4 Wp/m^2 for JW-HD156N.

Time representativeness:

The study used primary data collected from Jan. 2022 to Dec. 2022.

Database(s) and LCA software used:

In the context of this study, the activity data are mainly of 'primary type', i.e. collected with the support of the company for the specific production sites. Generic data related to the life cycle impacts of the material or energy flows that enter and leave the production system is sourced from Ecoinvent 3.9 "allocation, cut-off by allocation - unit" database. Secondary data such as silicon ingot and silicon wafer production are taken from IEA PVPS Task 12, 2020 report. LCA-software SimaPro version 9.5 was used.

Internal follow-up procedures:

In order to keep the LCA data representative and reliable, input data for the LCA model as well as information in the EPD, such as raw material acquisition, transportation modes, manufacturing processes, changes in product design etc. will be checked annually by Jolywood internally. If there would be any significant changes taking place, the LCA model, LCA report and EPD report would be updated accordingly and summited for review.

A4 Transport

PARAMETER	VALUE/DESCRIPTION
Fuel type and consumption of vehicle or vehicle type used for transport e.g. long distance truck, boat, etc.	Truck of 16- 32 ton: Fuel consumption: 0.014~0.015 L/100 km. Ship: Fuel consumption: 9.0E-4~9.6E-4 L/100 km
Distance	Ship: 8646.5 km Truck: 1141.9 km
Capacity utilization (including empty returns)	Truck: 36.7 % (full + empty return) Ship: 70 % (full + empty return)
Bulk density of transported products	N/A
Volume capacity utilization factor	1

A5 Installation

PARAMETER	VALUE/DESCRIPTION
Auxiliary materials for installation	N/A
Use of water	0
Use of other resources	N/A
Quantitative description of the type of energy (regional mix) and the consumption during the installation process	Electricity: 6.0E-5 kWh/FU Diesel: 3.0E-4 kg/FU
Wastage of materials on the building site before waste processing, generated by the product’s installation (specified by type)	Non-hazardous waste: 2.35E-3~2.5E-3 kg/FU

B Use stage

PARAMETER	VALUE/DESCRIPTION
Auxiliary materials for use phase	N/A
Use of water	1.82E-2~1.96E-2 kg/FU
Use of other resources	N/A
Quantitative description of the type of energy (regional mix) and the consumption during use	No energy consumption during use
Wastage of materials on the building site before waste processing, generated during use	No waste generation

C End of life stage

PARAMETER	VALUE/DESCRIPTION
Waste collection process specified by type	Non-hazardous waste: 5.48E-2~5.55E-2 kg/FU
Recovery system specified by type	Incineration with energy recovery: 3.97E-4~4.27E-4 kg/FU
Waste processing Recovery system specified by type Recovery system specified by type	Materials recycling: 4.17E-2~4.27E-2 kg/FU
Disposal Characteristic performance, Disposal specified by type	Inert materials landfill: 7.92E-3~8.10E-3 kg/FU

System diagram:

Table 5 LCA stages

System boundaries (X = included, MND = module not declared, MNR = module not relavant)																		
	Product Stage			Construction stage		Use Stage							End-of-life stage					Beyond the system boundaries
	Raw Material	Transport	Manufacturing	Transport	Assembly / Install	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	De-construction and demolition	Transport	Waste processing	disposal		Reuse-Recovery-Recycling-potential
Module	A1	A2	A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1	C2	C3	C4		D
Modules declared	X	X	X	X	X	ND	X	ND	ND	ND	ND	ND	X	X	X	X		X
Geography	CN			Global		-	-	-	-	-	-	-	-	GLO				GLO
Specific data used	>90%			-	-	-	-	-	-	-	-	-	-	-	-	-		-
Variation-products	<10%			-	-	-	-	-	-	-	-	-	-	-	-	-		-
Variation-sites	0%			-	-	-	-	-	-	-	-	-	-	-	-	-		-

System boundary of this study is cradle to gate with options, modules C1–C4, module D and with optional modules (A1–A3 + C + D and A4-A5, B2), as illustrated jn Figure 3.

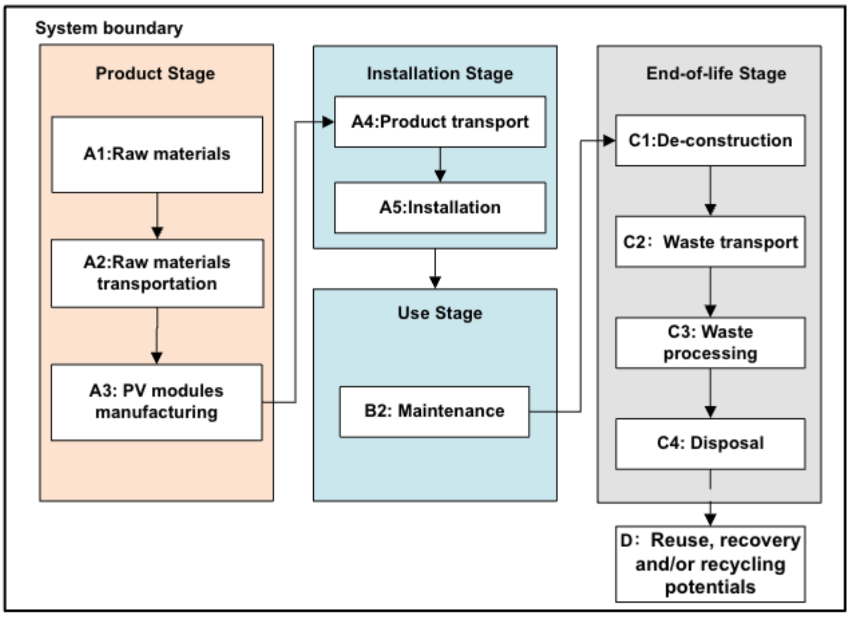


Figure 3 System boundary

Excluded Processes:

The following steps/stages are not included in the system boundary due to the reason that the elements below are considered irrelevant or not within the boundary to the LCA study:

- Impacts related to the production, transportation and installation of capital goods (buildings, infrastructure, machinery, internal transport packaging) and general operations (staff travel, marketing and communication actions) that cannot be directly allocated to products are excluded from the LCA study.
- The packaging for silicon wafer and solar cells is reused internally and its impact was excluded from the system;
- Emissions during the PV module installation and operation due to no obvious emission observable.

- Storage phases and sales of PV products due to no observable impact. Product losses due to abnormal damage such as natural disasters or fire accidents would occur at a rather low frequency.
- Handling operations at the distribution center and retail outlet due to small contribution and negligible impact.
- Research and development activities.
- Long-term emissions.

Assumptions:

In general, this LCA study is subject to certain assumptions and limitations, which have been made in a conservative manner following the LCA standard and related specifications listed in the PCR. It should be noted that caution should be taken when comparing the results of PV modules in this report with those from other studies. The assumptions made are listed below:

Table 6 List of assumptions

Categories	Items	Assumptions
Manufacturing stage (A1-A3)	Silicon wafer	Life cycle inventory (LCI) data of silicon ingot and the silicon wafer is difficult to obtain at the stage, thus an average LCI data for China in IEA PVPS Task 12,2020 is used for modelling
Transportation stage (A2 & A4)	Transportation vehicle type	For the vehicle used in raw materials and product transportation, CN6 type vehicles are used, which are aligned with the EURO6 standards, EURO 6 type vehicle with 16-32 ton capacity is assumed for modelling.
Installation stage (A5)	Electricity and materials use	The electricity and diesel consumption have been derived from the ecoinvent database process for a photovoltaic plant installation with a capacity of 570 kWp, and these values have been downscaled to the power output of different modules analysed in this study
Use & Maintenance	Use (B1)	The use stage requires no energy and materials inputs, and has no emissions.
	Maintenance (B2)	Water used for cleaning in 25 years is assumed with 0.23L per per module per time and two times per year (link)
	Replacement (B4)	No replacement for the module as the module has RSL>25 years
End-of-life (C1-C4)	De-construction (C1)	The de-construction of PV modules is assumed to be done manually, no electricity and materials use in this stage
	Waste transportation (C2)	Waste transportation distance from the de-installation plant to the waste treatment facilities is assumed to be 50 km for simplification purposes.
	Waste processing (C3)	The electricity consumption during this stage is 0.277kWh/kg module based on the data from IEA.
	Disposal (C4)	Disposal scenarios is based on the WEEE

Allocation:

The allocation is made in accordance with the provisions of PCR. Allocation refers to the partitioning of input or output flows of a process or a product system between the product systems under study and one or more other product systems. In this study, there are three types of allocation procedures considered:

Multi-input allocation

The allocation of electricity and emissions during the manufacturing stage of PV module are allocated by power output ratio. The transportation of raw materials is allocated by mass ratio.

Multi-output allocation

No other by-products are produced from the production, hence there is no production of by-products that need to be used to allocate the situation.

End-of-life allocation

For end-of-life allocation of background data (energy and materials), the model "allocation cut-off by classification (ISO standard) is used. The underlying philosophy of this approach is that primary (first) production of materials is always allocated to the primary user of a material. If material is recycled, the primary producer does not receive any credit for the provision of any recyclable materials. Consequently, recyclable materials are available burden-free for recycling processes, and secondary (recycled) materials bear only the impacts of the recycling processes.

For end-of-life stage of the solar PV module products, polluter-pays-principle (PPP) is followed. As for the load and benefit of reuse, recycling, and recovery processes (Module D), it is reported separately following the PCR's recommendation. End-of-life approach with 100/0 allocation is adopted in this analysis. This approach is based on the assumption that material recycled into secondary material at end-of-life substitutes an equivalent amount of virgin material (losses considered). Hence a credit is given to account for this material substitution. However, this also means that burdens equivalent to this credit should be assigned to scrap used as an input to the production process. which means that the environmental impact of the recycled material is the same as that of the virgin material. This approach rewards end-of-life recycling but does not reward the use of recycled content.

In cases where materials are sent to waste incineration, the study accounts for waste composition, heating value, and regional efficiencies, and assigns credits for power and heat outputs using the regional grid mix and thermal energy from natural gas, resulting in a conservative estimate of the benefits of energy recovery.

Cut-off rules:

The following procedure was followed for the exclusion of inputs and outputs:

- All inputs and outputs to a (unit) process will be included in the calculation for which data is available. Data gaps may be filled by conservative assumptions with average or generic data. Any assumptions for such choices will be documented;
- According to PCR, life cycle inventory data shall according to EN 15804 include a minimum of 95% of total inflows (mass and energy) per module. In addition, if less than 100% of the inflows are accounted for, proxy data or extrapolation should be used to achieve 100% completeness.

Electricity mix:

In this LCA study, it is important to note that different electricity grid mixes are used for different stages of the life cycle. Specifically, the production of solar cells and PV modules takes place in Jiangsu Province, China, where the Eastern China grid electricity mix is used. On the other hand, the end-of-life stage, which involves PV module waste disposal and recycling, taking European market as a case study, where the European average grid electricity mix is used.

It is worth noting that the use of different grid electricity mixes may impact the results of the LCA study, as the environmental impacts associated with electricity generation can vary significantly between different regions and grid mixes. Therefore, it is important to carefully consider the impact of grid mix differences when interpreting the results of the study, and to recognize that the results may not be directly comparable to studies that use different grid mixes.

Environmental performance

Results of JW-HD156N-182 (630W as representative results) are selected to cover the two products in this EPD. Note that the results are relative expressions and do not predict impacts on endpoint categories, exceedance of certain levels, safety margins or risks.

Table 7 Mandatory impact category indicators according to EN 15804

Indicator	Unit	A1-A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1	C2	C3	C4	D
GWP-fossil	kg CO_2 eq.	4.44E-01	1.69E-02	1.70E-03	ND	1.61E-05	ND	ND	ND	ND	ND	1.41E-03	5.19E-04	5.37E-03	8.72E-04	-9.38E-02
GWP-biogenic	kg CO_2 eq.	-1.37E-03	7.35E-06	2.97E-03	ND	9.59E-08	ND	ND	ND	ND	ND	6.12E-07	5.15E-07	2.03E-04	2.39E-06	9.05E-04
GWP- luluc	kg CO_2 eq.	4.30E-04	9.78E-06	3.18E-07	ND	2.51E-08	ND	ND	ND	ND	ND	2.44E-07	2.52E-07	1.32E-05	8.86E-08	-1.87E-04
GWP- total	kg CO_2 eq.	4.43E-01	1.69E-02	4.66E-03	ND	1.62E-05	ND	ND	ND	ND	ND	1.41E-03	5.20E-04	5.58E-03	8.74E-04	-9.30E-02
ODP	kg CFC11 eq	7.34E-09	2.87E-10	2.73E-11	ND	2.43E-12	ND	ND	ND	ND	ND	2.15E-11	1.10E-11	9.47E-11	1.88E-11	-1.16E-09
AP	mol H+ eq	2.54E-03	1.71E-04	1.32E-05	ND	8.38E-08	ND	ND	ND	ND	ND	1.26E-05	1.11E-06	2.65E-05	8.13E-07	-6.65E-04
EP-freshwater	kg P eq.	1.14E-04	1.06E-06	7.71E-08	ND	6.86E-09	ND	ND	ND	ND	ND	6.12E-08	3.60E-08	4.82E-06	3.44E-08	-4.15E-05
EP-marine	kg N eq	5.51E-04	4.26E-05	6.04E-06	ND	1.69E-08	ND	ND	ND	ND	ND	5.78E-06	2.79E-07	4.71E-06	2.97E-07	-1.24E-04
EP-terrestrial	mol N eq	5.46E-03	4.65E-04	6.54E-05	ND	1.71E-07	ND	ND	ND	ND	ND	6.28E-05	2.83E-06	4.15E-05	2.88E-06	-1.46E-03
POCP	kg NMVOC eq.	1.59E-03	1.46E-04	1.94E-05	ND	5.57E-08	ND	ND	ND	ND	ND	1.86E-05	1.72E-06	1.34E-05	1.07E-06	-4.03E-04
ADP-minerals & metals*	Kg Sb eq.	1.22E-05	4.27E-08	9.48E-10	ND	7.29E-11	ND	ND	ND	ND	ND	5.07E-10	1.65E-09	1.05E-08	5.29E-10	-7.87E-06
ADP-fossil*	MJ	4.99E+00	2.24E-01	1.98E-02	ND	2.01E-04	ND	ND	ND	ND	ND	1.81E-02	7.18E-03	1.21E-01	2.42E-03	-1.17E+00
WDP*	m^3	4.76E-01	8.59E-04	6.24E-05	ND	7.32E-04	ND	ND	ND	ND	ND	4.64E-05	3.00E-05	1.26E-03	9.45E-05	-1.61E-02
Acronyms	GWP-fossil = Global Warming Potential fossil fuels; GWP-biogenic = Global Warming Potential biogenic; GWP-luluc = Global Warming Potential land use and land use change; ODP = Depletion potential of the stratospheric ozone layer; AP = Acidification potential, Accumulated Exceedance; EP-freshwater = Eutrophication potential, fraction of nutrients reaching freshwater end compartment; EP-marine = Eutrophication potential, fraction of nutrients reaching marine end compartment; EP-terrestrial = Eutrophication potential, Accumulated Exceedance; POCP = Formation potential of tropospheric ozone; ADP-minerals&metals = Abiotic depletion potential for non-fossil resources; ADP-fossil = Abiotic depletion for fossil resources potential; WDP = Water (user) deprivation potential, deprivation-weighted water consumption															

* The results of this environmental impact indicator shall be used with care as the uncertainties of these results are high or as there is limited experience with the indicator.

Table 8 Additional environmental impacts

Indicator	Unit	A1-A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1	C2	C3	C4	D
GWP-GHG [1]	kg CO_2 eq.	4.45E-01	1.69E-02	1.70E-03	ND	1.61E-05	ND	ND	ND	ND	ND	1.41E-03	5.19E-04	5.38E-03	8.72E-04	-9.39E-02

[1] According to the PCR, a supplementary indicator for climate impact (GWP-GHG) shall be reported. This indicator includes all greenhouse gases excluding biogenic carbon dioxide uptake and emissions and biogenic carbon stored in the product as defined by IPCC AR 6 (IPCC 2021).

Use of resources

Table 9 Resource use

Indicator	Unit	A1-A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1	C2	C3	C4	D
PERE	MJ	9.49E-01	2.76E-03	3.30E-02	ND	2.14E-05	ND	ND	ND	ND	ND	1.70E-04	1.13E-04	2.33E-02	1.69E-04	-1.69E-01
PERM	MJ	3.28E-02	0.00E+00	-3.28E-02	ND	0.00E+00	ND	ND	ND	ND	ND	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
PERT	MJ	9.81E-01	2.76E-03	2.06E-04	ND	2.14E-05	ND	ND	ND	ND	ND	1.70E-04	1.13E-04	2.33E-02	1.69E-04	-1.69E-01
PENRE	MJ	5.61E+00	2.19E-01	1.93E-02	ND	2.13E-04	ND	ND	ND	ND	ND	1.75E-02	6.94E-03	5.64E-02	5.15E-01	-1.25E+00
PENRM	MJ	5.13E-01	0.00E+00	0.00E+00	ND	0.00E+00	ND	ND	ND	ND	ND	0.00E+00	0.00E+00	0.00E+00	-5.13E-01	0.00E+00
PENRT	MJ	6.12E+00	2.19E-01	1.93E-02	ND	2.13E-04	ND	ND	ND	ND	ND	1.75E-02	6.94E-03	5.64E-02	2.33E-03	-1.25E+00
SM	kg	0.00E+00	0.00E+00	0.00E+00	ND	0.00E+00	ND	ND	ND	ND	ND	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
RSF	MJ	0.00E+00	0.00E+00	0.00E+00	ND	0.00E+00	ND	ND	ND	ND	ND	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
NRSF	MJ	0.00E+00	0.00E+00	0.00E+00	ND	0.00E+00	ND	ND	ND	ND	ND	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
FW	m^3	1.17E-02	2.85E-05	2.52E-06	ND	1.76E-05	ND	ND	ND	ND	ND	1.68E-06	1.03E-06	9.50E-05	2.64E-05	-5.68E-04
Acronyms	PERE = Use of renewable primary energy excluding renewable primary energy resources used as raw materials; PERM = Use of renewable primary energy resources used as raw materials; PERT = Total use of renewable primary energy resources; PENRE = Use of non-renewable primary energy excluding non-renewable primary energy resources used as raw materials; PENRM = Use of non-renewable primary energy resources used as raw materials; PENRT = Total use of non-renewable primary energy re-sources; SM = Use of secondary material; RSF = Use of renewable secondary fuels; NRSF = Use of non-renewable secondary fuels; FW = Use of net fresh water															

Waste production and output flows

Table 10 Waste production

Indicator	Unit	A1-A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1	C2	C3	C4	D
HWD	kg	6.68E-04	1.35E-06	1.31E-07	ND	5.62E-10	ND	ND	ND	ND	ND	1.19E-07	4.57E-08	1.54E-07	1.28E-08	6.12E-06
NHWD	kg	3.91E-02	8.32E-03	1.16E-04	ND	2.32E-06	ND	ND	ND	ND	ND	2.77E-05	3.57E-04	4.68E-02	7.90E-03	-1.18E-02
RWD	kg	5.39E-06	4.92E-08	4.09E-09	ND	4.90E-10	ND	ND	ND	ND	ND	3.47E-09	2.36E-09	8.84E-07	2.24E-09	-2.24E-06
Acronyms	HWD = Hazardous waste disposed; NHWD = Non-hazardous waste disposed; RWD = Radioactive waste disposed															

Table 11 Output flows

Indicator	Unit	A1-A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1	C2	C3	C4	D
CRU	kg	0.00E+00	0.00E+00	0.00E+00	ND	0.00E+00	ND	ND	ND	ND	ND	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
MFR	kg	6.25E-04	0.00E+00	1.64E-03	ND	0.00E+00	ND	ND	ND	ND	ND	0.00E+00	0.00E+00	4.17E-02	0.00E+00	0.00E+00
MER	kg	1.03E-05	0.00E+00	1.31E-03	ND	0.00E+00	ND	ND	ND	ND	ND	0.00E+00	0.00E+00	3.97E-04	0.00E+00	0.00E+00
EEE	MJ	0.00E+00	0.00E+00	0.00E+00	ND	0.00E+00	ND	ND	ND	ND	ND	0.00E+00	0.00E+00	0.00E+00	0.00E+00	2.43E-03
EET	MJ	0.00E+00	0.00E+00	0.00E+00	ND	0.00E+00	ND	ND	ND	ND	ND	0.00E+00	0.00E+00	0.00E+00	0.00E+00	1.35E-03
Acronyms	CRU = Components for re-use; MFR = Materials for recycling; MER = Materials for energy recovery; EEE = Exported electrical energy; EET = Exported thermal energy															

Information related to EPD of multiple products

Results of JW-HD144N-182 (585W) in this report are presented by its A1-A3 stage results and variation against JW-HD156N-182 (630W) results.

Table 12 Environmental impacts of JW-HD144N-182 (585W) and variations

Parameter	Unit	A1-A3	Variations, %
GWP-fossil	kg CO_2 eq.	4.18E-01	-5.9%
GWP-biogenic	kg CO_2 eq.	-9.56E-04	-30.0%
GWP- luluc	kg CO_2 eq.	3.96E-04	-7.9%
GWP- total	kg CO_2 eq.	4.17E-01	-5.8%
ODP	kg CFC11 eq	6.85E-09	-6.7%
AP	mol H+ eq	2.41E-03	-5.2%
EP-freshwater	kg P eq.	1.07E-04	-6.1%
EP-marine	kg N eq	5.22E-04	-5.6%
EP-terrestrial	mol N eq	5.19E-03	-5.2%
POCP	kg NMVOC eq.	1.51E-03	-5.2%
ADP-minerals & metals*	Kg Sb eq.	1.27E-05	3.7%
ADP-fossil*	MJ	4.69E+00	-6.0%
WDP*	m^3	4.69E+00	-6.7%

Note: 1. Negative variations indicates lower results;

2. A significant variation in GWP-biogenic is attributed to different amount of packaging material used.

** Disclaimer: The results of this environmental impact indicator shall be used with care as the uncertainties of these results are high or as there is limited experience with the indicator.*

Table 13 Additional environmental impacts of JW-HD144N-182 (585W) and variations

Parameter	Unit	A1-A3	Variations, %
GWP-GHG[1]	kg CO_2 eq.	4.18E-01	-5.9%

[1] According to the PCR, a supplementary indicator for climate impact (GWP-GHG) shall be reported. This indicator includes all greenhouse gases excluding biogenic carbon dioxide uptake and emissions and biogenic carbon stored in the product as defined by IPCC AR 6 (IPCC 2021).

Table 14 Resource use of JW-HD144N-182 (585W) and variations

Indicators		Unit	A1-A3	Variations, %
Primary energy resources – Renewable	Used as energy carrier	MJ, net calorific value	9.49E-01	-6.9%
	Used as raw materials	MJ, net calorific value	3.28E-02	-19.6%
	TOTAL	MJ, net calorific value	9.81E-01	-7.4%
Primary energy resources – Non-renewable	Used as energy carrier	MJ, net calorific value	5.61E+00	-6.0%
	Used as raw materials	MJ, net calorific value	5.13E-01	-9.0%
	TOTAL	MJ, net calorific value	6.12E+00	-6.2%
Secondary material		kg	0.00E+00	0.0%
Renewable secondary fuels		MJ, net calorific value	0.00E+00	0.0%
Non-renewable secondary fuels		MJ, net calorific value	0.00E+00	0.0%
Net use of fresh water		m^3	1.17E-02	-6.8%

Table 15 Waste production of JW-HD144N-182 (585W) and variations

Indicators	Unit	A1-A3	Variations, %
Hazardous waste disposed	kg	6.68E-04	-6.9%
Non-hazardous waste disposed	kg	3.91E-02	-6.5%
Radioactive waste disposed	kg	5.39E-06	-6.3%

Table 16 Output flows of JW-HD144N-182 (585W) and variations

Indicators	Unit	A1-A3	Variations, %
Components for reuse	kg	0.00E+00	0.0%
Material for recycling	kg	6.25E-04	0.6%
Materials for energy recovery	kg	1.03E-05	0.4%
Exported electrical energy	MJ	0.00E+00	0.0%
Exported thermal energy	MJ	0.00E+00	0.0%

JW-HD156N-182 (630W) has been chosen as representative products for its series. To obtain results across various power output ranges for the products, a list of conversion factors has been provided, accounting for different peak power ranges.

Table 17 EPD results conversion factors of various power output ranges for JW-HD156N-182 series

Rated power output range (Wp)	605	610	615	620	625
Conversion factor	1.041	1.033	1.024	1.016	1.008

Interpretation of results

Manufacturing stage, specifically the raw materials extraction stage, is responsible for most impacts. Three key materials that contribute the most to the environmental impacts in PV module production are solar cells, glass, and frame.

Additional environmental information

Radiology, Electro Magnetic Fields and Noise

The impact of photovoltaic power generation is mainly the noise of power generation equipment such as inverters and transformers, the low-level radiation generated by photovoltaic cells, and electromagnetism. The noise is within a controllable range and below the legally set limits. The radiation has little impact on the human body and the environment, similar to the battery of mobile phones and cameras. The electromagnetic field and the radio and TV interferences that will be generated by the operation of the connection line will not exceed the recommended limits. For the end-of-life stage of the photovoltaic power station, the battery components need to be recycled. In China, crystalline silicon battery components are widely used, therefore producing very little pollution.

Environmental risks

The manufacturing of silicon material and silicon wafer is the most energy-consuming part in the production process of the entire industrial chain. During the production of crystalline silicon, a large amount of silicon tetrachloride ($SiCl_4$) and dichloro-dihydro silicon (SiH_2Cl_2) are produced, and these by-products can be treated through resource reuse. The recycling process is strictly controlled.

During the manufacturing of solar cells and photovoltaic modules. The pollution comes mainly from the cell production process. Chemicals like hydrofluoric, hydrochloric acid, isopropanol, hydrogen peroxide, etc. are used, and the discharge of exhaust gas will also affect the environment. Jolywood strictly controls the discharge of waste water and exhaust gas, and has been awarded the “National green factory” title by the Ministry of Industry and Information Technology of the Chinese government. Assembling the solar cells into photovoltaic modules does not consume chemicals and basically does not cause pollution to the environment.

Energy source of manufacturing process and its climate impact

Electricity provider	Unit	GWP-GHG
Eastern China electricity grid mix (Electricity, medium voltage {CN-ECGC}| market for electricity, medium voltage | Cut-off, U)	kg CO_2-eq/kWh	0.887

Information related to sectorial EPD

This EPD is not sectorial.

Differences with previous versions

Representative results were declared to present different products. Conversion factors were added.

References

LCA for Jolywood PV modules

R. Frischknecht, P. Stolz, L. Krebs, M. de Wild-Scholten, P. Sinha, V. Fthenakis, H. C. Kim, M. Raugei, M. Stucki, 2020, Life Cycle Inventories and Life Cycle Assessment of Photovoltaic Systems, International Energy Agency (IEA) PVPS Task 12, Report T12-19:2020.

INTERNATIONAL EPD SYSTEM

General Programme Instructions of the International EPD® System. Version 4.0.

PCR 2019:14 Construction Products, Version 1.3.1

c-PCR-016 Photovoltaic modules and parts thereof (adopted from EPD Norway 2022-04-27)

INTERNATIONAL AND EUROPEAN STANDARDS

EN 15804:2012+A2:2019/AC:2021 Sustainability of construction works - Environmental product declarations - Core rules for the product category of construction products

ISO 14020:2022 Environmental statements and programmes for products: Principles and general requirements

ISO 14025:2011 Environmental labels and declarations - Type III environmental declarations -Principles and procedures

ISO 14040: 2006/Amd 1:2020 Environmental management - Life cycle assessment - Principles and framework Amendment 1 (ISO 2020)

ISO 14044: 2006/Amd 2:2020 Environmental management - Life cycle assessment - Requirements and guidelines Amendment 2 (ISO 2020)

WEEE Directive 2012/19/EU Article 4,11&15

LCA SOFTWARE AND DATABASE

SimaPro 9.5, LCA software

Ecoinvent Database 3.9

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